A Barnes integral for the fixed-mass recoil graph

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Abstract

We reduce the full internal double-sine integral of the continued tetrahedron graph to a Barnes integral, at fixed positive edge masses in the pure-recoil limit. The transported contour becomes a horizontal line separating two pole families. Exact q-Gamma factorization and a uniform integrable tail bound justify passage of the limit through this integral. Its vertical-contour representation yields four convergent ${}_4F_3(1)$ terms for distinct pole families, while the integral remains regular when they coalesce. Combining this result with the previous vertex limit gives normalized fixed-topology graph ratios. Physical external-energy cycles, insertion normalization and the joint heavy-mass saddle limit remain uncomputed. This is an author self-reviewed draft, without independent certification or a claim of priority.

1 Labels, contour and the exact change of variable

Take the de Sitter radius to be one and use CLP [1], (3.46). The six edge masses are $\mu_j > 0$. Put

$$b^2 = i/\kappa, \quad L = b^{-2} = -i\kappa, \quad Q = b + b^{-1}, \qquad iP_j = -\frac{1}{2b} - \frac{ib\mu_j}{2}.$$

Let

$$\mathcal{V} = \{123, 156, 246, 345\}, \qquad \mathcal{R} = \{1245, 1346, 2356\}, \qquad a_v = \frac{1}{2}\sum_{j\in v}\mu_j, \quad r_\rho = \frac{1}{2}\sum_{j\in\rho}\mu_j.$$

Each edge occurs twice in both collections, so $\sum_v a_v = \sum_\rho r_\rho$. Assume the strict separation

$$a_{\max} < r_{\min}, \qquad c = (a_{\max} + r_{\min})/2. \tag{1}$$

All masses and this gap are fixed as $\kappa \to \infty$.

In the internal integral set

$$iP = -\frac{3}{2b} + bz, \qquad \mathrm{d}P = -ib\,\mathrm{d}z. \tag{2}$$

Its exact integrand, before the measure, is

$$K_\kappa(z) = \frac{\prod_v S_b(b(z + ia_v))}{\prod_\rho S_b(b(z + \frac{1}{2} + ir_\rho + L))\, S_b(b(z + \frac{1}{2} - L))}. \tag{3}$$

The original numerator pole families and denominator-zero families become, with $n, m \geq 0$,

$$z = -n + i\kappa m - ia_v, \tag{4}$$

$$z = \tfrac{1}{2} + n - i\kappa m - ir_\rho, \qquad z = \tfrac{1}{2} + n - i\kappa(m+2). \tag{5}$$

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Cancellations may remove poles; these lists contain all possible poles.

CLP (3.58), at the endpoint, gives $s = \mathbb{R} + \pi/\kappa - 8\pi i + 2\pi i(a_{\max} + r_{\min})/\kappa$. Using $s = -8\pi i - 4\pi(z - \frac{1}{2})/\kappa$, its image is exactly

$$\mathcal{C}_z = \mathbb{R} - ic. \tag{6}$$

For sufficiently large κ, (1) puts the first families above and the latter families below this line. We orient $\mathcal{C}_z$ left to right and use the corresponding transported orientation in (2); reversing both changes only an overall sign. This is an internal contour, not an external three-energy contour for a physical correlator.

2 Exact q-Gamma factorization

Write $q = e^{-2\pi/\kappa}$, $\tau = e^{-2\pi\kappa}$, and define

$$A_v = z + ia_v, \quad D_\rho = z + \tfrac{1}{2} + ir_\rho, \quad D_0 = z + \tfrac{1}{2}, \qquad \Delta_2 = \sum_v (A_v^2 - A_v) - \sum_{\rho,0} (D_\rho^2 - D_\rho).$$

The last sum includes all four D's. Then

$$\sum_v A_v - \sum_{\rho,0} D_\rho = -2, \qquad \Delta_2 = -4z + \sum_\rho r_\rho^2 - \sum_v a_v^2 + 1 - i\sum_\rho r_\rho.$$

CLP (A.5) and the q-Gamma definition [2] give exactly

$$S_b(bw) = C_\kappa (1-q)^{w-1} e^{i\pi w/2 + \pi(w^2-w)/(2\kappa)} \Gamma_q(w) (\tau e^{2\pi i w}; \tau)_\infty, \tag{7}$$

where $C_\kappa = (q;q)_\infty^{-1} e^{-i\pi(Q^2+1)/12}$ is independent of w. The inverse-b shift is $S_b(bw + b^{-1}) = 2\sin(\pi w) S_b(bw)$. Apply it once to each of the three shifted denominators and inversely to the fourth. The normalized exact integrand is

$$\mathcal{B}_\kappa(z) = -8\pi^3 e^{-\pi\kappa} (1-q)^2 K_\kappa(z), \tag{8}$$

$$= \frac{\pi^3 e^{-i\pi z + \pi\Delta_2/(2\kappa)} (1 + \tau e^{2\pi i z})}{\prod_\rho \sin(\pi D_\rho)} \frac{\prod_v \Gamma_q(A_v)}{\prod_{\rho,0} \Gamma_q(D_\rho)} \frac{\prod_v (\tau e^{2\pi i A_v}; \tau)_\infty}{\prod_{\rho,0} (\tau e^{2\pi i D_\rho}; \tau)_\infty}. \tag{9}$$

No small-b approximation has been made in this formula. On compact subsets avoiding poles, the ordinary $q \to 1^-$ limit, reflection formula and the small-τ products yield

$$\mathcal{B}_\kappa(z) \longrightarrow \mathcal{B}(z) := e^{-i\pi z} \frac{\prod_v \Gamma(z + ia_v) \prod_\rho \Gamma(\frac{1}{2} - z - ir_\rho)}{\Gamma(z + \frac{1}{2})}. \tag{10}$$

3 An integrable bound for both tails

Compact convergence alone would not establish the integral limit. Here the mass balance supplies the missing tail control.

For $q = e^{-h}$, put $f_q(x) = \log \Gamma_q(x)$ at real $x > 0$. Differentiating the absolutely convergent product and comparing its positive sum with an integral gives

$$\log \frac{1-q^x}{1-q} - \frac{h}{e^{hx} - 1} \le f_q'(x) \le \log \frac{1-q^x}{1-q}, \tag{11}$$

$$|f_q''(x+s)| \le \frac{h}{e^{h(x-A)} - 1} + \frac{h^2 e^{h(x-A)}}{(e^{h(x-A)} - 1)^2} \le \frac{1}{x - A} + \frac{1}{(x-A)^2}, \tag{12}$$

provided $\operatorname{Re} s \ge -A$ and $x > A$. The complex bound uses $|1 - q^{x+s+n}| \ge 1 - q^{x-A+n}$.

Thus, for any two fixed collections of four shifts with difference of sums -2, Taylor expansion along the shifts gives, uniformly for large x and $0 < h \le h_0$,

$$\left|\frac{\prod_{j=1}^{4}\Gamma_q(x+\alpha_j)}{\prod_{j=1}^{4}\Gamma_q(x+\delta_j)}\right| \le C\left(\frac{1-q}{1-e^{-hx}}\right)^2. \tag{13}$$

The constant depends on the shifts. In (9), on $z = x - ic$, all sine denominators stay away from zero, and the modular products and their inverses are bounded for sufficiently large κ. The quadratic exponential contributes e^{-hx}. Consequently its positive tail is bounded by a constant times

$$e^{-hx}\left(\frac{1-e^{-h}}{1-e^{-hx}}\right)^2 \le \left(\frac{h}{2\sinh(hx/2)}\right)^2 \le x^{-2}.$$

For clarity the negative tail is not inferred from the positive one. Reflect all eight original double sines via $S_b(Q-x) = 1/S_b(x)$ and put $w = -z$. The exact integrand becomes

$$K_\kappa(-w) = \frac{\prod_\rho S_b(b(w+\frac{1}{2}-ir_\rho))\, S_b(b(w+\frac{1}{2}+2L))}{\prod_v S_b(b(w+1-ia_v+L))}. \tag{14}$$

After inverse-b shifts its four small numerator arguments have offsets $\frac{1}{2} - ir_\rho, \frac{1}{2}$, and its four denominator offsets are $1 - ia_v$. Their sum difference is again -2. The shift factors are

$$\frac{[2\sin\pi(w+\frac{1}{2}+L)][2\sin\pi(w+\frac{1}{2})]}{\prod_v 2\sin\pi(w+1-ia_v)}.$$

On $w = x + ic$, the denominator imaginary parts are $c - a_v > 0$. Removing $e^{\pi\kappa}$ leaves a bounded trigonometric factor, and the same argument proves the negative-tail bound. Hence

$$|\mathcal{B}_\kappa(x-ic)| \le \frac{C}{1+x^2} \quad (\kappa \ge \kappa_0,\ x \in \mathbb{R}). \tag{15}$$

On the remaining compact interval use (9) and compact convergence. The constants here depend on the fixed masses and gap; no joint growing-mass bound is claimed.

Proposition 1. *Under (1), the normalized internal integral satisfies*

$$-\frac{4\pi^2(1-q)^2e^{-\pi\kappa}}{b}\,\mathcal{J}_b \longrightarrow \mathfrak{B}(\boldsymbol{\mu}) := \frac{1}{2\pi i}\int_{\mathbb{R}-ic}\mathcal{B}(z)\,\mathrm{d}z = \frac{1}{2\pi i}\int_{d-i\infty}^{d+i\infty}\mathcal{B}(z)\,\mathrm{d}z, \quad 0 < d < \tfrac{1}{2}. \tag{16}$$

Proof. Dominated convergence follows from (15); the normalization uses $\mathrm{d}P = -ib\,\mathrm{d}z$ and the definition of $\mathcal{B}_\kappa$. For the second equality rotate the positive horizontal tail upwards and the negative tail downwards about $d - ic$. No poles enter either swept quadrant. Gamma reflection and fixed-shift asymptotics give $O(|z|^{-2})$ on the connecting arcs, with exponential improvement away from their horizontal ends, so the arc contributions vanish. The vertical contour separates $z = -n - ia_v$ from $z = \frac{1}{2} + n - ir_\rho$. Its two tails decay as $e^{-2\pi y}$ for $y \to +\infty$ and $e^{-4\pi|y|}$ for $y \to -\infty$, times powers of $|y|$. □

4 A hypergeometric evaluation and coincident families

For distinct a_v, close the vertical contour to the left and sum the four pole families. Standard Barnes residue calculus [3] gives

$$\mathfrak{B} = \sum_{k=1}^{4} c_k\, {}_4F_3\left(\begin{matrix}\frac{1}{2}+ia_k,\ \{\frac{1}{2}+i(a_k-r_\rho)\}_{\rho=1}^{3} \\ \{1+i(a_k-a_j)\}_{j\ne k}\end{matrix}; 1\right), \tag{17}$$

$$c_k = e^{-\pi a_k}\frac{\prod_{j\ne k}\Gamma(i(a_j-a_k))\prod_\rho\Gamma(\frac{1}{2}+i(a_k-r_\rho))}{\Gamma(\frac{1}{2}-ia_k)}. \tag{18}$$

The sum of lower parameters minus the sum of upper parameters is one, so each series converges. For example $\Gamma(A-n) = (-1)^n\Gamma(A)/(1-A)_n$; all six signs from the residue, phase, other left factors and reciprocal denominator cancel. Closing arcs can be chosen away from the pole rows, where the balanced quotient is $O(|z|^{-2})$.

Individual coefficients become singular when some $a_j = a_k$. The integral (16) is regular there and is the definition to use, or one may take the limit of the complete sum. Assigning values to the divergent summands separately is invalid. Barnes/Racah representations have substantial antecedents; we make no novelty claim or identification with a particular unitary group $6j$ without its representation and measure dictionary.

5 What this now determines, and what it does not

Let $T(\mu_i, \mu_j, \mu_k)$ be the limiting combined vertex from the preceding note. For a reference six-mass configuration $\boldsymbol{\mu}^\star$ satisfying (1) and $\mathfrak{B}(\boldsymbol{\mu}^\star) \neq 0$, we obtain the full fixed-topology graph ratio

$$\lim_{\kappa\to\infty} \frac{Z_{\mathrm{tet}}(\boldsymbol{\mu};\kappa)}{Z_{\mathrm{tet}}(\boldsymbol{\mu}^\star;\kappa)} = \prod_{v\in\mathcal{V}} \frac{T(\boldsymbol{\mu}_v)}{T(\boldsymbol{\mu}_v^\star)} \left|\frac{\mathfrak{B}(\boldsymbol{\mu})}{\mathfrak{B}(\boldsymbol{\mu}^\star)}\right|^2 . \tag{19}$$

This is a graph result at fixed real masses, not a physical four-point normalization prescription.

To connect to the prior transverse coordinates, choose explicitly

$$(\mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6) = (m_1, \nu, m_2, m_3, \nu, m_4).$$

Then the four a_v's are the four values $M + \nu/2 + (\pm u \pm v)/2$, independent of ω; the three r_ρ's are $M + \nu + \omega/2$, $2M$, $M + \nu - \omega/2$. At $u = v = 0$ all four left pole families coalesce, so (17) is not a stable termwise formula. This explicit labeling may reverse the sign of ω relative to CLP's symmetric diagram; the earlier Gram expressions are even in that sign.

The next calculable step is the large-M, ν saddle analysis of (16), retaining u, v, with the full vertex factor. Its internal determinant must be derived in this representation before comparison with the previous full-$6j$ Gram hypothesis. The tail constant in (15) has not been made uniform in growing masses or complex external energies. Analytic continuation of (19) requires separate chiral continuation, not a new modulus at complex masses. The physical insertion factors, external energy cycles and disconnected baseline remain undetermined. Finite-G dynamics and a complete de Sitter dual remain open.

6 Numerical checks

The script `code/recoil_internal.py` records 34 passing checks: four symbolic balances, nine exact shift identities and precision refinements, three limiting integrals at 45/65 digits, one hypergeometric comparison, and eight two-sided tail refinements. The four-term sum has cancellation ratio 2.68×10^5 in its sample; coalesced families are evaluated by the integral. Relative convergence errors are saved separately from arithmetic agreement. These are not interval certificates or direct full finite-κ quadratures. The uniform tail result uses the analytic argument above.

References

[1] J. Chandra, D. Liska and B. Post, *3D de Sitter from complex Virasoro TQFT: Observers and knots*, arXiv:2610.30247v1, (3.46), (3.48), (3.52), (3.58), Appendix A.

[2] NIST DLMF, Section 5.18, q-Gamma product and classical limit; read online.

[3] NIST DLMF, Section 16.5, Barnes representations and residue continuation; read online.